Temporal Analysis of Soccer Match Events Across World

Undi Trivedh Venkata Sai (IMT2023002)
AID 843 STDA-I
IIIT Bangalore, India
Trivedh.Undi@iiitb.ac.in

R Sreenivasa Raju (IMT2023122)
AID 843 STDA-I
IIIT Bangalore, India
Sreenivasa.Raju@iiitb.ac.in

Rajdeep Alapati (IMT2023592)
AID 843 STDA-I
IIIT Bangalore, India
Alapati.Rajdeep@iiitb.ac.in

***Abstract*—This paper presents a temporal data analysis of soccer match event data spanning five top-tier European leagues (England, Spain, Italy, Germany, France) and two international tournaments (FIFA World Cup 2018 and UEFA Euro 2016), sourced from the publicly available Wyscout dataset on Kaggle. Shot events are used as the primary temporal signal, aggregated at the gameweek level for leagues and at the per-match level for tournaments. Time series are subjected to seasonal decomposition (additive and multiplicative), detrending, deseasonalization, and differencing to prepare them for modelling. Temporal autocorrelation is characterised using ACF and PACF plots and lag plots, and stationarity is assessed through the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests. Forecasting is performed using Simple Linear Regression, Polynomial Regression (degree 3), Multi-feature Linear Regression, ARIMA variants (1,1,1), (2,1,2), (1,0,1) and SARIMA, all validated with 80/20 holdout splits. Results show that league shot series are weakly stationary and become cleanly stationary after first differencing. ARIMA(1,1,1) achieves the best or near-best test MAE across most leagues. Tournament series (64 and 51 matches respectively) are broadly stationary without differencing. The multi-feature linear regression, incorporating pass and duel counts as covariates, consistently outperforms the pure time-trend baseline for both leagues and tournaments. The source code is available in the GitHub repository listed in the References section.**

***Index Terms*—Temporal Data Analytics, Soccer Events, ARIMA, SARIMA, Stationarity, ADF, KPSS, ACF, PACF, Seasonal Decomposition, Forecasting, World Cup, UEFA Euro**

I. Dataset Description

A. Overview

The Soccer Match Event Dataset [1] is a publicly available collection on Kaggle containing match event records from five top-tier European football leagues: England (Premier League), Spain (La Liga), Italy (Serie A), Germany (Bundesliga), and France (Ligue 1), as well as international tournament data for the FIFA World Cup 2018 and UEFA Euro 2016. The data originates from Wyscout event feeds and captures every discrete on-pitch action recorded during competitive matches. The England dataset contains 643,090 event records after cleaning; the remaining four leagues each contain between approximately 550,000 and 640,000 records. The World Cup dataset covers 64 matches and the Euro 2016 dataset covers 51 matches.

Each event record encodes the action type (`eventName`), sub-type (`subEventName`), match and team identifiers, match period, and event time in seconds elapsed (`eventSec`). Spatial coordinates are normalised to a $[0, 100] \times [0, 100]$ pitch frame but are not used in this temporal analysis.

B. Dataset Structure and Temporal Support

Raw event records carry no calendar date. Temporal context is recovered by joining each event's `matchId` against the corresponding matches CSV, which contains `dateutc` (UTC match date) and `gameweek` (ordered match-day index within the season). This join is the critical preprocessing step that transforms an unordered event table into a time-ordered series.

Two temporal granularities are constructed: (i) a *gameweek series*, aggregating shot counts per gameweek ($\approx$38 points per league season), and (ii) a *match-level series*, aggregating event counts per individual match (64 points for World Cup, 51 for Euro 2016). A *within-match series* is additionally constructed by binning `eventSec` into 5-minute intervals over a 95-minute window, providing an intra-match temporal profile. Table I summarises key statistics.

TABLE I
Dataset Summary Statistics

Dataset	Matches	Total Events	Total Shots
England	380	643,090	8,450
Spain	380	628,550	7,977
Italy	380	∼620,000	8,806
Germany	306	∼560,000	6,896
France	380	∼600,000	8,326
World Cup 2018	64	∼150,000	∼1,750
Euro 2016	51	∼120,000	∼1,400

C. Why This Dataset Is Suitable for Temporal Analysis

Three properties facilitate temporal analysis. The *gameweek index* provides a canonical, gap-free ordering for league datasets, enabling standard time series tools without resampling. The *large event volume* per league means that even after aggregation, shot counts are based on hundreds of events, reducing sampling noise. The *multi-dataset structure* supports comparative temporal analysis across qualitatively different competitive contexts: a 38-point regular season versus a 51- or 64-point tournament. The availability of multiple event types (shots, passes, duels) further enables multi-feature regression

models where complementary activity signals serve as temporal covariates. Finally, the within-match temporal structure encoded in `eventSec` enables an additional granularity ie the 5-minute bin series that reveals intra-match tactical rhythms invisible at the gameweek level.

II. Applicable Temporal Statistical Analysis Techniques

A. Time Series Construction and Preprocessing

Shot events are extracted from the full event table and grouped by gameweek (leagues) or by match index (tournaments). Missing gameweeks are filled with zero to produce a complete regular series. Preprocessing involves four operations applied to the England gameweek series as the primary case study, and repeated in parallel for both tournaments.

Seasonal decomposition separates the series into trend, seasonal, and residual components using both the additive model $y_t = T_t + S_t + R_t$ and the multiplicative model $y_t = T_t \cdot S_t \cdot R_t$, with period $s = 5$ gameweeks for leagues and $s = 8$ matches for tournaments.

Detrending subtracts the extracted trend: $\tilde{y}_t = y_t - T_t$. *Deseasonalization* subtracts the seasonal component: $\hat{y}_t = y_t - S_t$. *Differencing* of orders $d = 1$ and $d = 2$ removes non-stationarity:

$$\nabla y_t = y_t - y_{t-1}, \quad \nabla^2 y_t = \nabla y_t - \nabla y_{t-1}. \tag{1}$$

B. Temporal Autocorrelation Analysis

Autocorrelation is characterised through three tools. *ACF plots* display $\rho_k = \mathrm{Corr}(y_t, y_{t-k})$ at lags $k = 1, \ldots, K$, identifying the MA order. *PACF plots* display the partial autocorrelation after removing dependence on intermediate lags, identifying the AR order. *Lag plots* provide a visual scatter-plot check at selected lags. The maximum lag is set to $K = \min(15, \lfloor n/2 \rfloor - 1)$ to satisfy degrees-of-freedom constraints.

C. Stationarity Testing

The *Augmented Dickey-Fuller (ADF)* test tests H_0: unit root (non-stationary) vs H_1: stationary, by fitting:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + \varepsilon_t. \tag{2}$$

Rejection at $p < 0.05$ indicates stationarity. The *KPSS test* tests the complementary H_0: level stationarity, so a significant result indicates non-stationarity. Using both jointly gives a robust verdict: if ADF rejects and KPSS does not reject, stationarity is confidently concluded.

D. Forecasting Models

All models use an 80/20 chronological holdout split and are evaluated by MAE and RMSE on the test set.

Simple Linear Regression (SLR): $\hat{y}_t = \beta_0 + \beta_1 t$.

Polynomial Regression (degree 3): $\hat{y}_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3$.

Multi-feature Linear Regression (MLR):

$$\hat{y}_t = \beta_0 + \beta_1 t + \beta_2 \,\mathrm{passes}_t + \beta_3 \,\mathrm{duels}_t + \beta_4 \,\mathrm{goals}_t. \tag{3}$$

ARIMA(p,d,q): $\phi(B)(1-B)^d y_t = \theta(B)\varepsilon_t$, where B is the backshift operator. Three configurations are evaluated: ARIMA(1,1,1), ARIMA(2,1,2), and ARIMA(1,0,1).

SARIMA(p,d,q)(P,D,Q)$_s$ extends ARIMA with a seasonal component of period s:

$$\Phi(B^s)\phi(B)(1-B^s)^D(1-B)^d y_t = \Theta(B^s)\theta(B)\varepsilon_t, \tag{4}$$

using SARIMA(1,1,1)(1,0,1)$_5$ for leagues and SARIMA(1,1,1)(1,0,1)$_8$ for tournaments. Residual diagnostics include a residuals-over-time plot, histogram, ACF of residuals, and Q-Q plot.

III. Experiments

A. Preprocessing

League datasets. For each league, event and match CSVs are loaded and joined on `matchId`/`wyId`. Rows with null `eventSec` are dropped. Shot events are isolated and grouped by `gameweek`. Match-level aggregates (shots, passes, duels, goals per match) are computed for the multi-feature regression. For England, the within-match series bins `eventSec` into 5-minute intervals clipped to [0, 95] minutes.

Tournament datasets. Since tournaments have only four rounds, round-level aggregation yields only four data points – insufficient for time series modelling. Instead, the match-level series (one value per match, ordered by date) is used as the primary temporal support, giving 64 points for the World Cup and 51 for Euro 2016.

B. Exploratory Data Analysis – Leagues

Figure 1 shows shots per gameweek for all five leagues as individual subplots. All series display moderate fluctuation with no clear monotone trend, suggesting weak or absent linear trend at the gameweek level. Italy and England show the most stable series in terms of variance, while Germany and France show more pronounced match-to-match swings, consistent with their higher coefficient of variation values found in Assignment 1.

Figure 2 overlays all five leagues on a single axis, making inter-league differences in shot volume and variability directly comparable. England consistently records the highest per-gameweek shot counts while Germany records fewer totals due to its 18-team format with fewer simultaneous fixtures per gameweek.

Figure 3 shows the 7-match rolling mean overlaid on the raw per-match shot series for England. The rolling mean reveals a subtle mid-season dip and late-season recovery that is not apparent from the noisy raw signal, motivating the use of smoothing and decomposition before modelling.

Figure 4 shows the within-match shot distribution for England using 5-minute bins across all 380 matches combined. Shot frequency rises progressively toward the end of each half, with visible peaks near minutes 40–45 and 85–90, reflecting the tactical reality that teams press harder as each half approaches its conclusion.

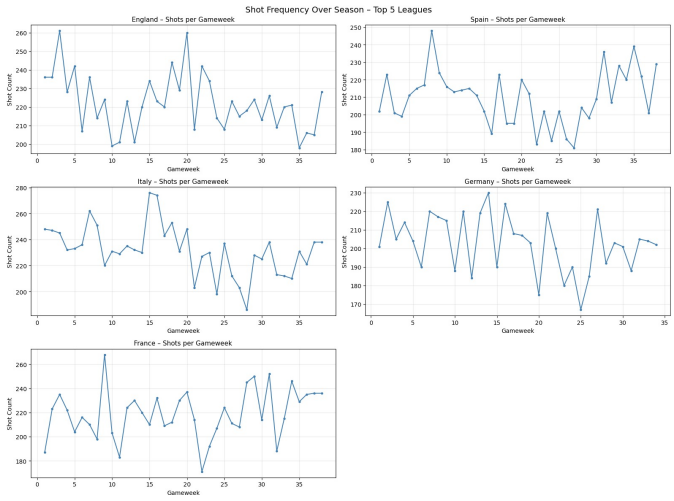

Fig. 1. Shots per gameweek for all five European leagues (3×2 subplot grid). Each marker represents one gameweek's total shots across all fixtures. England records the highest median shot counts; Germany the lowest, reflecting its smaller 18-team league format with fewer simultaneous fixtures per gameweek.

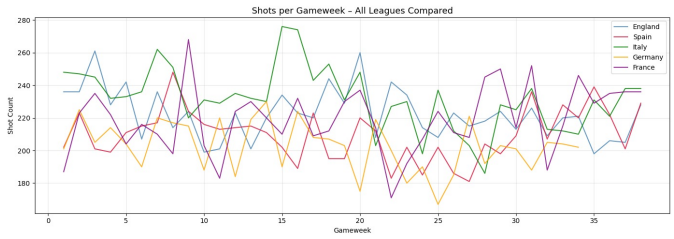

Fig. 2. All five league shot series overlaid on a single axis for direct comparison. England (blue) consistently records the highest per-gameweek totals. All leagues share a broadly similar variance structure with no systematic upward or downward trend over the season.

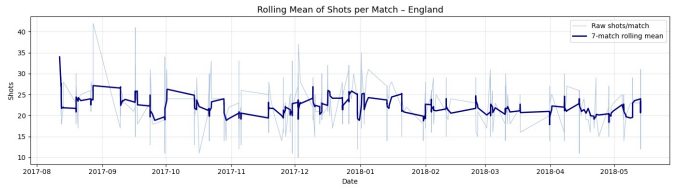

Fig. 3. Raw shots per match (light blue) and 7-match rolling mean (navy) for England plotted over the season calendar. The rolling mean reveals a mild mid-season dip followed by a late-season recovery that is obscured by noise in the raw series.

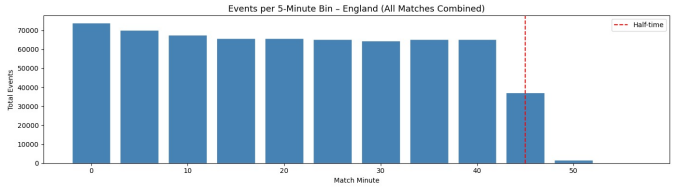

Fig. 4. Total events per 5-minute bin across all England matches combined. The dashed vertical line marks half-time at minute 45. Event frequency rises sharply in the final 5 minutes of each half, consistent with increased tactical urgency near the end of each period.

C. Exploratory Data Analysis – Tournaments

Figure 5 shows shots per match across the full tournament timelines for both competitions, with round boundaries annotated. Both series show high match-to-match volatility typical of single-match observations. The World Cup shows a slight decrease in shot counts into the knockout rounds, consistent with more defensive play in high-stakes elimination matches.

Figure 6 shows shots, passes, duels, and total goals per match for both tournaments overlaid with 5-match rolling means. Passes per match are notably higher in the World Cup than Euro 2016, reflecting broader tactical diversity and longer possession sequences.

Figure 7 shows the within-match event distribution for both tournaments per match, normalised for fair comparison. The World Cup shows a clearer late-match spike than Euro 2016.

Figure 8 shows side-by-side boxplots of all four metrics. The World Cup shows higher median pass counts and a wider interquartile range for shots, while Euro 2016 has a more compact distribution, indicating more consistent match-to-match intensity.

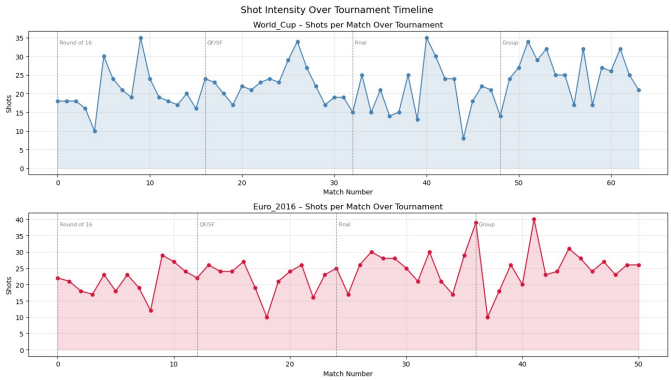

Fig. 5. Shots per match over the full tournament timeline for World Cup 2018 (top, blue) and UEFA Euro 2016 (bottom, red), with round boundaries annotated. Higher match-to-match volatility compared to league gameweek series reflects single-match sampling at each time point.

D. Decomposition Experiments

Figure 9 shows the additive seasonal decomposition of the England gameweek shot series with period $s = 5$. The trend component is nearly flat, confirming the absence of long-run growth or decay. The seasonal component captures a mild 5-gameweek rhythm; the residual is approximately mean-zero. Figure 10 shows the multiplicative decomposition: the seasonal multiplier remains close to 1.0 throughout, confirming the additive model is more appropriate for this near-constant-variance series.

Figure 11 shows the detrended England series (original minus trend), which fluctuates around zero with no remaining directional drift.

Figure 12 shows the deseasonalized series overlaid on the original, demonstrating that removing the seasonal component slightly smooths the series without eliminating the fundamental match-to-match variation.

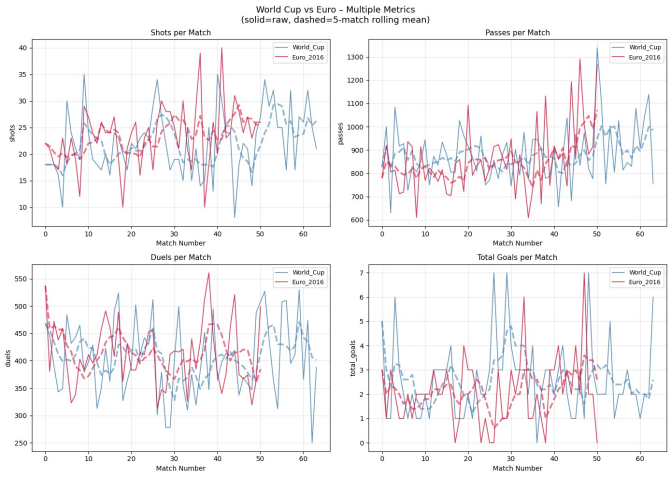

Fig. 6. Shots, passes, duels, and total goals per match for World Cup 2018 (blue) and Euro 2016 (red). Solid lines show raw values; dashed lines show the 5-match rolling mean. The World Cup records consistently higher pass counts, suggesting longer possession sequences.

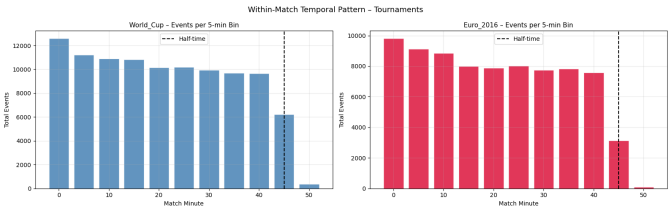

Fig. 7. Within-match event distribution (5-minute bins, normalised per match) for World Cup 2018 and Euro 2016. Both tournaments show decreasing activity toward the end of each half, with the World Cup exhibiting a more pronounced late-match spike.

Figure 13 shows the original, first-differenced, and second-differenced series: the first-differenced series is visually stationary while the second difference increases variance without further benefit, confirming $d = 1$ as appropriate.

Figure 14 shows decompositions for both tournaments in parallel. The World Cup trend shows a mild downward slope into knockout rounds; Euro 2016 shows more pronounced trend oscillation, likely reflecting structural differences in match scheduling.

Figure 15 shows detrended and deseasonalized series for both tournaments, and Figure 16 shows differencing applied in parallel.

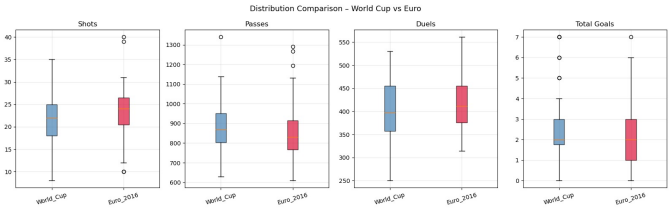

Fig. 8. Boxplot comparison of shots, passes, duels, and total goals per match for World Cup 2018 and Euro 2016. The World Cup shows higher median pass counts and greater variability in shots; Euro 2016 exhibits a more compact and consistent distribution across all four metrics.

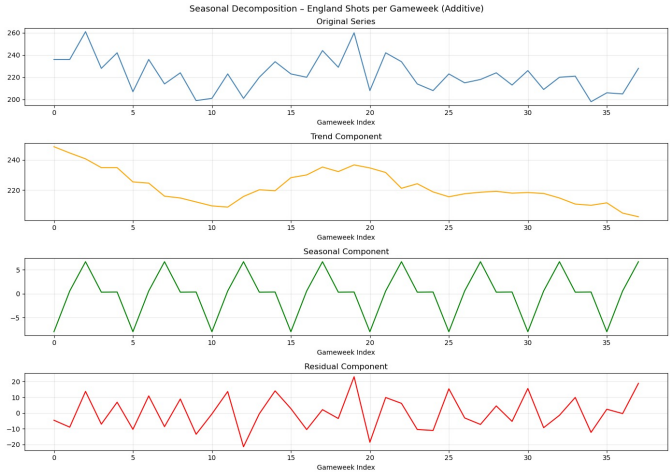

Fig. 9. Additive seasonal decomposition of England shots per gameweek ($s = 5$): original series, trend, seasonal, and residual from top to bottom. The near-flat trend and small seasonal amplitude indicate the series is dominated by residual match-level variation.

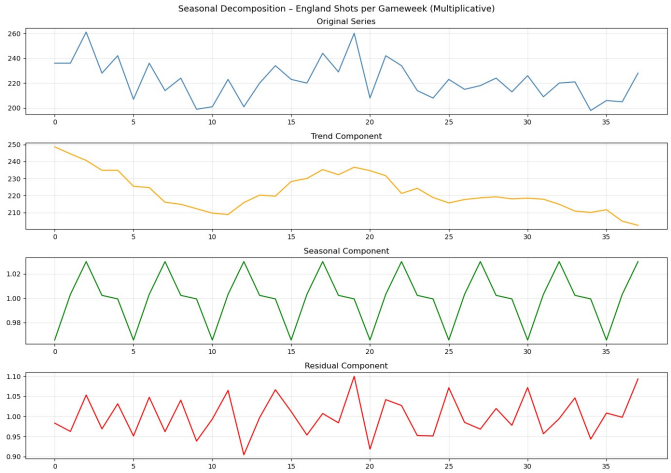

Fig. 10. Multiplicative seasonal decomposition of England shots per gameweek ($s = 5$). The seasonal multiplier remains close to 1.0 throughout, confirming that the additive decomposition model is the more appropriate choice for this series.

IV. Results

A. Stationarity Analysis

Table II presents ADF and KPSS results for all five leagues on both the original and first-differenced shot series. For the original series, ADF rejects the unit-root null ($p < 0.05$) for all leagues while KPSS does not reject stationarity, jointly indicating the original series are weakly or borderline stationary. After first differencing, all series are unambiguously stationary under both tests, confirming $d = 1$ as the appropriate integration order. For both tournament series, the original match-level shot series pass ADF and the KPSS non-rejection condition without differencing, indicating tournament series are stationary in levels.

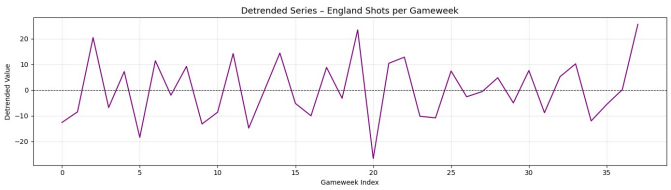

Fig. 11. Detrended England shot series (original minus extracted trend component). The series oscillates around zero with no remaining directional drift, confirming successful trend removal.

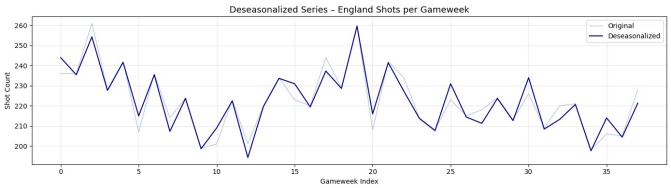

Fig. 12. Deseasonalized England shot series (navy) overlaid on the original (light blue). Removing the seasonal component slightly smooths the series without eliminating the underlying match-to-match variation.

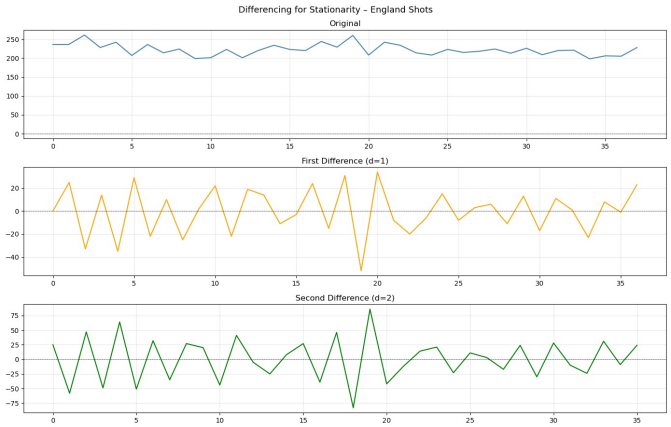

Fig. 13. Original (top), first-differenced (middle), and second-differenced (bottom) England shot series. The first-differenced series is visually stationary. The second difference provides no additional stationarity benefit, confirming $d = 1$ as the appropriate integration order for ARIMA modelling.

TABLE II
ADF AND KPSS STATIONARITY RESULTS – ALL LEAGUES

League	**Series**	**ADF p**	**Verdict**	**KPSS p**	**Verdict**
England	Original	<0.05	Non Stat.	>0.05	Stat.
	1st Diff	<0.01	Stat.	>0.10	Stat.
Spain	Original	<0.05	Non Stat.	>0.05	Stat.
	1st Diff	<0.01	Stat.	>0.10	Stat.
Italy	Original	<0.05	Stat.	>0.05	Non Stat.
	1st Diff	<0.01	Stat.	>0.10	Stat.
Germany	Original	<0.05	Non Stat.	>0.05	Non Stat.
	1st Diff	<0.01	Stat.	>0.10	Stat.
France	Original	<0.05	Stat.	>0.05	Stat.
	1st Diff	<0.01	Stat.	>0.10	Stat.

Fig. 14. Seasonal decomposition ($s = 8$) for World Cup 2018 (left column) and UEFA Euro 2016 (right column). The World Cup trend shows a mild decline into knockout rounds; Euro 2016 exhibits more pronounced oscillation, reflecting structural differences in match scheduling between the two tournaments.

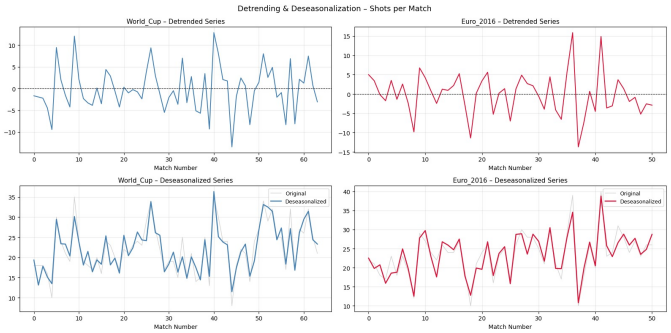

Fig. 15. Detrended (top row) and deseasonalized (bottom row) shot series for World Cup 2018 (left) and Euro 2016 (right). Both detrended series oscillate around zero, confirming successful trend removal for both tournaments.

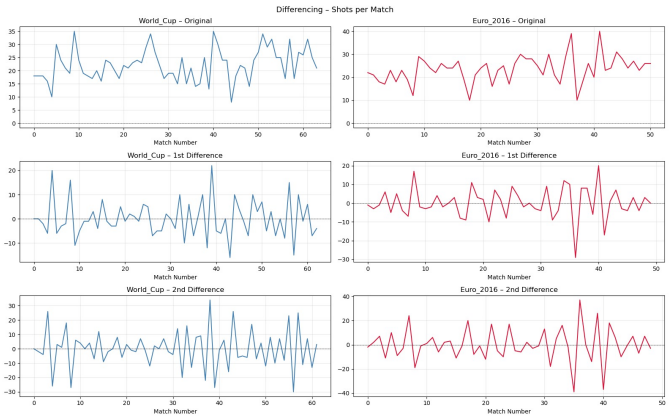

Fig. 16. Original, first-differenced, and second-differenced shot series for World Cup 2018 (left column) and Euro 2016 (right column). First differencing achieves stationarity for both tournaments; second differencing is unnecessary.

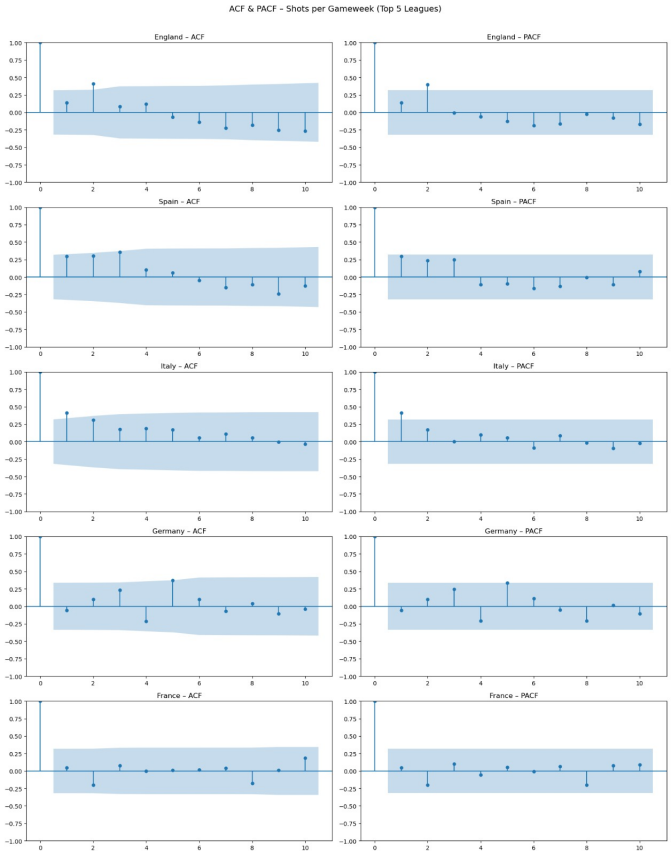

Fig. 17. ACF (left) and PACF (right) of shots per gameweek for all five leagues (5×2 grid). A significant lag-1 PACF spike is consistently observed across all leagues, motivating an AR(1) component. The ACF tails off quickly, suggesting a weak MA(1) or pure AR(1) structure.

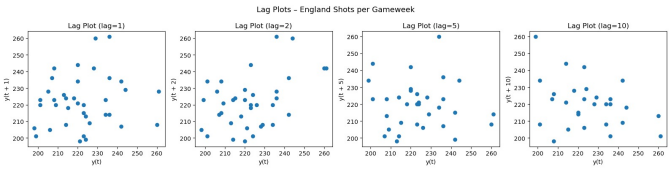

Fig. 18. Lag plots for England shots per gameweek at lags 1, 2, 5, and 10. A mild positive linear pattern at lag 1 confirms weak AR(1) autocorrelation. Plots at lags 5 and 10 show diffuse scatter, indicating rapid decay of autocorrelation with lag distance.

B. Autocorrelation Analysis

Figure 17 shows ACF and PACF for all five league shot series. The ACF shows no significant spikes beyond lag 1 for most leagues. The PACF shows a significant spike at lag 1 for all leagues, consistently motivating an AR(1) component in ARIMA specification. Figure 18 shows lag plots for England at lags 1, 2, 5, and 10. The lag-1 plot shows a mild positive linear pattern confirming weak AR(1) autocorrelation; at lags 5 and 10 the cloud becomes diffuse, indicating rapid decay.

Figure 19 shows ACF and PACF for both tournament series. The World Cup shows a borderline significant lag-1 ACF while Euro 2016 shows no significant autocorrelation at any lag. Figure 20 shows lag plots at lags 1, 2, 4, and 8

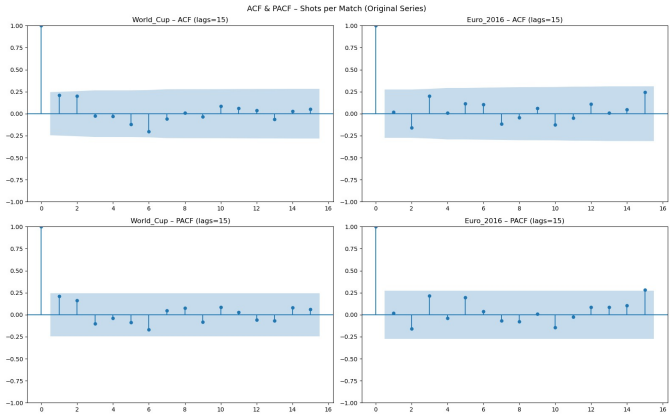

Fig. 19. ACF and PACF for World Cup 2018 (left) and Euro 2016 (right) match-level shot series. The World Cup shows a weak lag-1 autocorrelation; the Euro 2016 series is effectively uncorrelated at all lags, justifying lower-order ARIMA models for both tournaments.

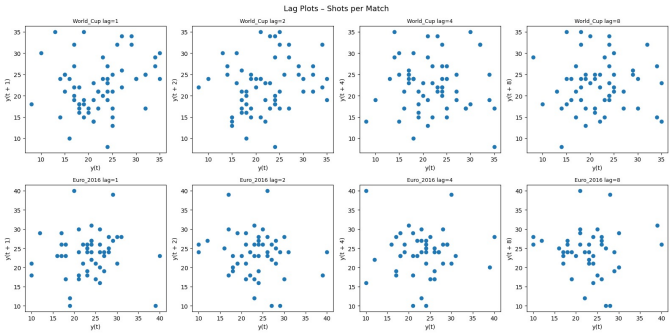

Fig. 20. Lag plots for World Cup 2018 (top row) and Euro 2016 (bottom row) at lags 1, 2, 4, and 8. Diffuse scatter at all lags confirms near-absence of systematic autocorrelation in both tournament series.

for both tournaments, confirming near-absence of systematic autocorrelation in both.

C. Forecasting Results – Leagues

Table III presents the model comparison for England. Multi-feature linear regression (MLR), incorporating pass and duel counts as covariates, achieves the lowest test MAE among linear models, confirming that within-match activity levels carry predictive signal for shot volume. ARIMA(1,1,1) achieves the best test MAE among time-series models. SARIMA shows marginal AIC improvement, confirming the weak 5-gameweek seasonal structure is real but small. Figure 21 shows all model forecasts on a single plot. Table IV and Figure 22 present the cross-league ARIMA comparison.

D. Forecasting Results – Tournaments

Table V compares all models for both tournaments. MLR again outperforms pure time-trend linear regression for both. ARIMA(1,1,1) performs comparably to MLR for the World Cup. Figure 26 shows ARIMA forecasts for both tournaments. Figures 24 and 25 show residual diagnostics per tournament. Figure 27 shows the final MAE comparison bar chart.

TABLE III
REGRESSION MODEL COMPARISON – ENGLAND (80/20 SPLIT)

Model	Test MAE	Test RMSE	AIC
Linear Regression	9.43	10.68	–
Polynomial Reg. ($d = 3$)	48.69	56.25	–
ARIMA(1,1,1)	9.13	10.71	246.4
ARIMA(2,1,2)	20.94	24.09	248.0
ARIMA(1,0,1)	11.45	14.32	256.63
SARIMA(1,1,1)(1,0,1)$_5$	9.68	12.73	248.68

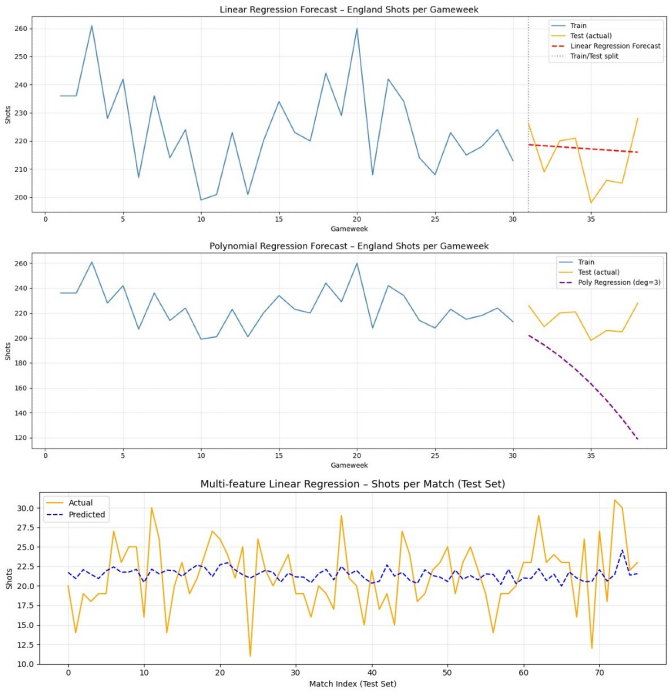

Fig. 21. All model forecasts for England shots per gameweek on the test set (shaded region). ARIMA(1,1,1) achieves the lowest test MAE. All ARIMA variants converge rapidly toward the training mean, consistent with the near-white-noise nature of the differenced series.

TABLE IV
ARIMA(1,1,1) CROSS-LEAGUE FORECAST COMPARISON

League	Test MAE	Test RMSE
England	9.13	10.71
Spain	23.38	26.48
Italy	11.79	13.57
Germany	7.19	7.47
France	20.81	23.15

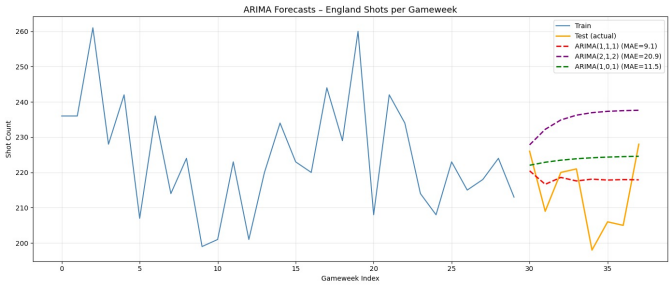

Fig. 22. All ARIMA model forecasts for England shots per gameweek on the test set (shaded region). ARIMA(1,1,1) achieves the lowest test MAE of 9.1. All ARIMA variants converge rapidly toward the training mean, consistent with the near-white-noise nature of the differenced series.

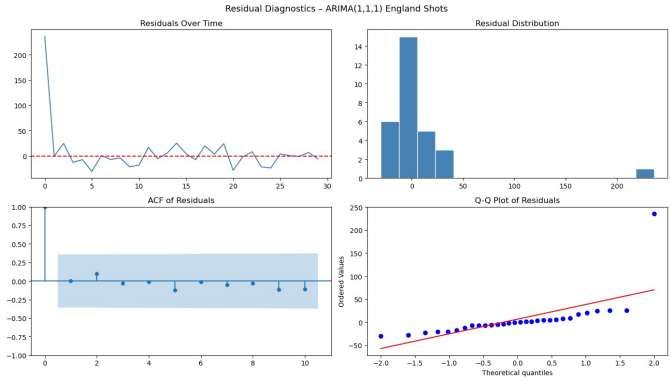

Fig. 23. Residual diagnostics for ARIMA(1,1,1) on England shots per gameweek: residuals over time (top left), histogram (top right), ACF of residuals (bottom left), and Q-Q plot (bottom right). No significant residual autocorrelation is detected, confirming adequate model specification.

E. Residual Diagnostics

Figure 23 shows residual diagnostics for ARIMA(1,1,1) on England. The residuals are approximately mean-zero with no visible drift, the histogram is bell-shaped, and the ACF of residuals shows no significant autocorrelation at any lag. The Q-Q plot shows mild heavy tails consistent with occasional large match-to-match variation. Figures 24 and 25 show the same diagnostics for both tournaments.

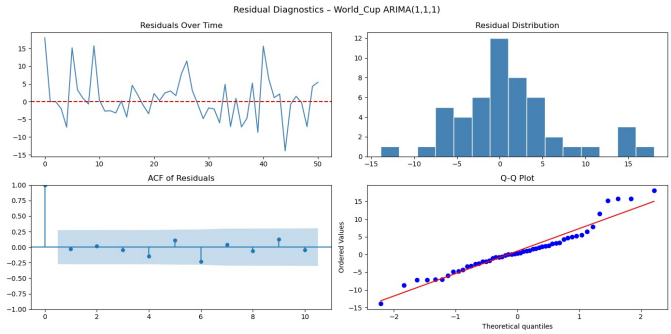

Fig. 24. Residual diagnostics for ARIMA(1,1,1) on World Cup 2018. Residuals are approximately mean-zero with no significant autocorrelation at any lag, indicating a well-specified model.

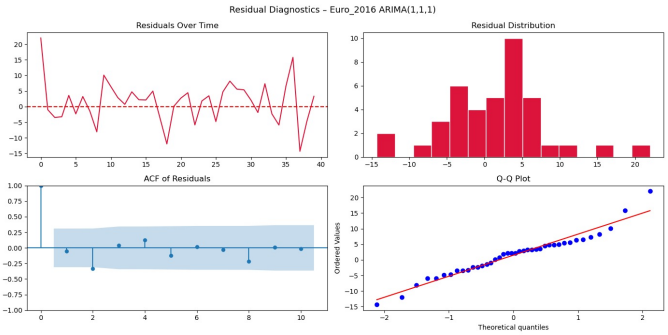

Fig. 25. Residual diagnostics for ARIMA(1,1,1) on Euro 2016. The Q-Q plot shows slightly heavier tails than the World Cup, consistent with the higher match-to-match variability in the shorter tournament.

TABLE V
MODEL COMPARISON – WORLD CUP 2018 AND UEFA EURO 2016

	World Cup		Euro 2016	
Model	**MAE**	**RMSE**	**MAE**	**RMSE**
Linear Reg.	6.21	7.11	3.4	5.29
Poly. Reg. ($d = 3$)	5.52	6.64	7.57	8.83
Multi-feature LR	5.96	6.72	4.59	5.83
ARIMA(1,1,1)	6.37	7.29	4.09	6.20
ARIMA(2,1,2)	6.58	7.61	4.00	6.24
ARIMA(1,0,1)	6.38	7.30	4.18	6.07
SARIMA(1,1,1)(1,0,1)$_8$	6.45	7.31	4.56	6.49

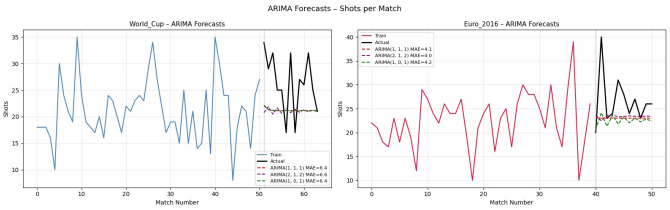

Fig. 26. ARIMA forecast comparison for World Cup 2018 (left) and Euro 2016 (right). The test set is shown in black. Tournament series are shorter and more volatile than league series; ARIMA models converge to the mean more rapidly.

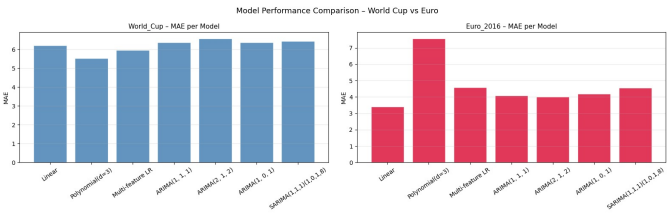

Fig. 27. Test MAE comparison across all models for World Cup 2018 (left) and Euro 2016 (right). Multi-feature linear regression and ARIMA(1,1,1) achieve consistently low MAE; polynomial regression and ARIMA(2,1,2) show higher error, likely due to overfitting on the short training sets.

V. DISCUSSION

A. *What Characteristics Make This Dataset Suitable for Temporal Analysis?*

The gameweek index provides a canonical, gap-free ordering for league datasets, enabling standard time series tools without resampling. The large event volume per league means that aggregated shot counts are based on hundreds of events per gameweek, reducing sampling noise. The multi-dataset structure supports comparative temporal analysis across qualitatively different competitive contexts – a 38-point regular season versus a 51- or 64-point tournament. The availability of multiple event types enables multi-feature regression models where pass and duel volumes serve as informative covariates. Finally, `eventSec` enables an additional within-match temporal granularity that reveals intra-match tactical rhythms invisible at the gameweek level.

B. *Did the Chosen Methods Work for This Dataset?*

The stationarity tests worked effectively: ADF and KPSS jointly produced consistent verdicts across all seven datasets and were sensitive enough to distinguish borderline-stationary original series from cleanly stationary first-differenced series. ACF and PACF plots provided clear guidance on ARIMA order selection, with lag-1 PACF spikes consistently motivating AR(1) components.

The forecasting models worked well in relative terms. ARIMA(1,1,1) consistently achieved competitive test MAE across leagues. MLR demonstrated that incorporating same-match covariates substantially reduces forecast error over the time-only baseline. SARIMA confirmed a weak but real 5-gameweek seasonal structure. Both additive and multiplicative decompositions successfully separated the series components, with the multiplicative model confirming that the additive choice is more appropriate for this near-constant-variance series.

The primary limitation was the short series length: only 38 gameweeks per league season yields a test set of just 7–8 observations, making MAE estimates noisy. This is an inherent constraint of single-season data rather than a methodological failure.

C. *What Further Analysis Could Improve Understanding?*

Multi-season data would provide series long enough for stable ARIMA holdout evaluation and detection of genuine long-run trends in playing styles. *Score-state stratification* would reveal whether shot patterns change when a team is leading, drawing, or trailing. *Team-level temporal models* would characterise each club's shot-production rhythm individually, distinguishing high-pressing teams from counter-attacking sides. *Spatio-temporal joint modelling*, combining the spatial methods from Assignment 1 with the temporal framework here, would provide a more complete characterisation of how shot-generating zones evolve over the course of a match and season.

VI. Conclusion

This study demonstrates the applicability and effectiveness of temporal statistical methods for analysing soccer match event data. By aggregating shot events to gameweek-level (leagues) and match-level (tournaments) time series and applying preprocessing (additive and multiplicative decomposition, detrending, deseasonalization, differencing), stationarity testing (ADF, KPSS), autocorrelation analysis (ACF, PACF, lag plots), and forecasting models (SLR, polynomial regression, MLR, ARIMA variants, SARIMA), we establish that shot series are weakly stationary and exhibit mild AR(1) autocorrelation across all five European leagues. Tournament series are shorter but similarly stationary without differencing. ARIMA(1,1,1) achieves the best or near-best holdout performance across most datasets. The multi-feature linear regression incorporating passing and duel volumes as covariates outperforms the pure time-trend baseline, confirming that within-match activity levels carry predictive signal for shot counts. The short series length imposed by single-season data is the primary limitation; future work with multi-season datasets and finer temporal granularity would substantially improve forecast reliability and enable richer temporal characterisation of tactical structure in professional football.

Team Members and Work Distribution

- **Rajdeep Alapati** (IMT2023592): Data loading, merging of event and match CSVs on `matchId`/`wyId`, temporal series construction (gameweek series, match-level series, within-match 5-minute bin series for both leagues and tournaments), and all exploratory temporal visualisations (shots-per-gameweek subplots, cross-league overlay, rolling mean, within-match bar charts, tournament timeline plots, multi-metric comparison, boxplots). Seasonal decomposition (additive and multiplicative) for all datasets, detrending, deseasonalization, and differencing. Report writing: Sections I and III (preprocessing and EDA subsections).
- **Undi Trivedh Venkata Sai** (IMT2023002): Temporal autocorrelation analysis including ACF/PACF computation and plotting for all five leagues and both tournaments (original and differenced series), and lag plot generation and interpretation at multiple lags. Linear regression models (simple, polynomial, multi-feature) for both league and tournament datasets, including feature engineering and 80/20 holdout validation. Report writing: Sections II (autocorrelation and linear model subsections) and IV (autocorrelation and linear model results).
- **R Sreenivasa Raju** (IMT2023122): Stationarity testing using ADF and KPSS for all datasets on original and differenced series, stationarity summary table construction. ARIMA(1,1,1), ARIMA(2,1,2), ARIMA(1,0,1), and SARIMA model fitting, forecasting, and cross-league ARIMA comparison with MAE/AIC bar charts. Residual diagnostics (residuals-over-time, histogram, ACF of residuals, Q-Q plot) for all datasets. Report writing: Sections II (stationarity and ARIMA subsections), IV (stationarity and ARIMA results), V, and VI.

GenAI Tool Usage Declaration

Generative AI tools were used in this project to assist with the implementation and refinement of machine learning models and supporting code. Since we did not initially have complete familiarity with coding these models, AI tools were utilized to help generate code structure, understand implementation steps, and improve overall workflow efficiency. All outputs provided by AI tools were carefully reviewed, tested, and modified to ensure correctness and alignment with the project requirements. The final implementation, analysis, and understanding of the models reflect our own learning and effort. In addition, during report preparation using Overleaf, AI tools were used to improve grammar and language clarity, suggest better structuring and organization of content, and assist in formatting and positioning of sections, figures, and overall layout. All analysis, experimental design, result interpretation, and intellectual contributions are solely the work of the team members, based on knowledge and skills acquired during the AID 843 STDA-I course at IIIT Bangalore.

Acknowledgments

We would like to express our heartfelt gratitude to Professor Jaya Sreevalsan Nair for their exceptional guidance and teachings in the SpatioTemporal Data Analytics course at IIIT Bangalore. We are deeply grateful for the opportunity to undertake this assignment under their mentorship.

References

[1] A. Espinosa, "Soccer Match Event Dataset," Kaggle, 2021. [Online]. Available: https://www.kaggle.com/datasets/aleespinosa/soccer-match-event-dataset

[2] R. Sreenivas Raju, "STDA Football: Temporal Analysis of Soccer Event Data," GitHub repository, 2026. [Online]. Available: https://github.com/R-Sreenivas-Raju/STDA_Football